

Condition-Gated Reasoning for Context-Dependent Biomedical Question Answering

Jash Parekh · Wonbin Kweon · Joey Chan · Rezarta Islamaj · Robert Leaman · Pengcheng Jiang · Chih-Hsuan Wei · Zhizheng Wang · Zhiyong Lu · Jiawei Han

University of Illinois Urbana-Champaign · National Institutes of Health



Motivation

Nearly every clinical decision is **conditional** — the right drug, test, or dose depends on the patient's history. However, existing Knowledge-Graph LLM biomedical QA systems assume medical knowledge applies uniformly and have the following limitations:

Static Knowledge

Knowledge graphs store bare facts: (entity, relation, entity), and treat every edge as always true; there is no slot for the conditions under which a fact holds.

Relevance $\neq$ Validity

Retrieval ranks by semantic similarity, so contraindicated facts that only look relevant might propagate to the answer. One new patient fact can invalidate an otherwise valid recommendation.

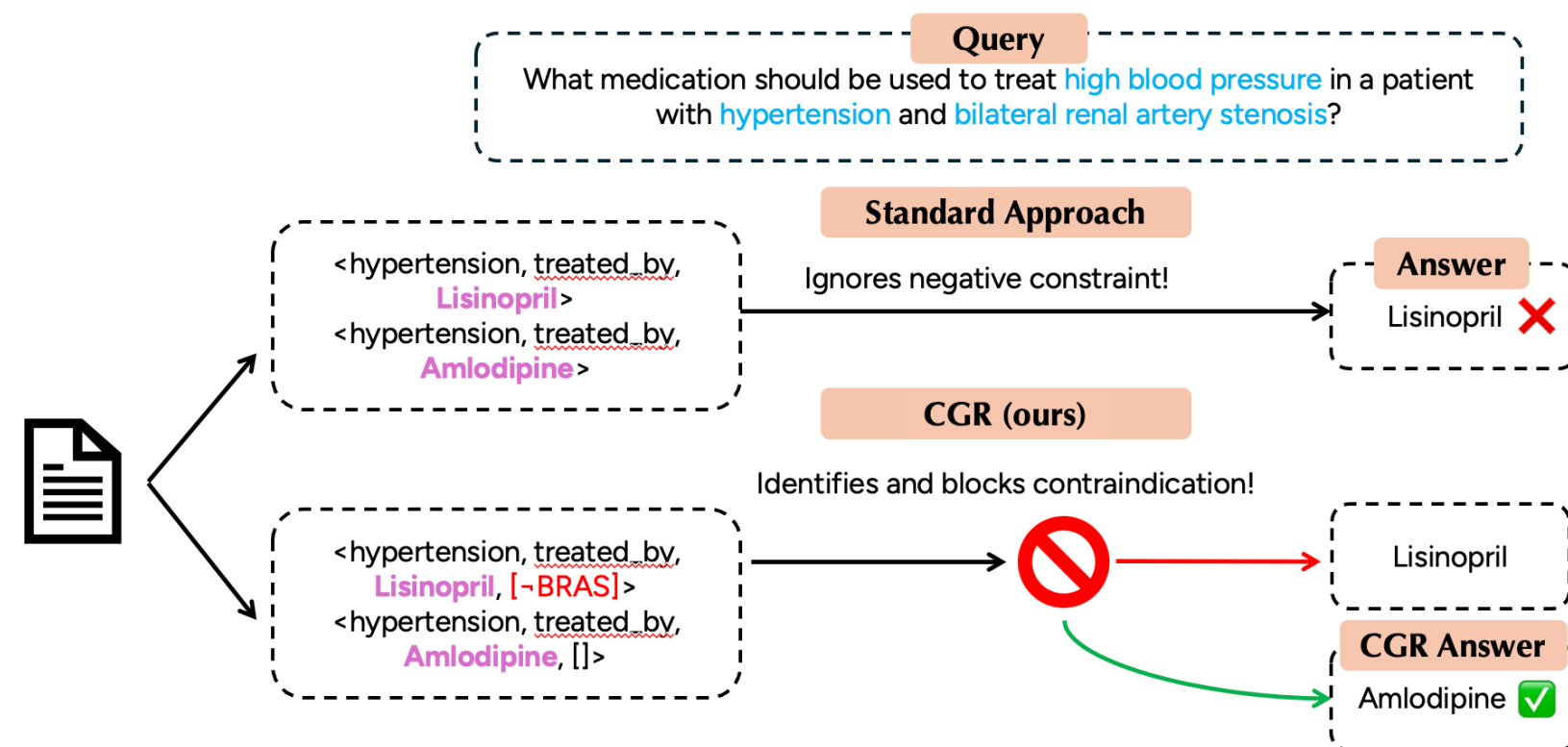
Non-monotonic Reasoning

One new patient fact can overturn an otherwise correct recommendation — a reversal retrieve-and-generate pipelines cannot represent.

OUR THESIS

Conditionality must be represented **explicitly** — a first-class property of knowledge, not something the model is made to infer.

EXAMPLE



A patient with hypertension *and* bilateral renal artery stenosis: the standard KG surfaces the contraindicated **Lisinopril**; CGR gates it out and returns the safe **Amlodipine**.

Condition-Gated Reasoning

CGR makes contextual conditions **integral parts of KG edges**, then **gates traversal** on the compatibility between each edge's conditions and the query — pruning contraindicated paths before they reach the answer.

Condition-aware extraction using LLM

4-tuples (**entity, relation, entity, conditions**); UMLS-normalized; evidence kept on each edge.

Single-pass condition evaluation

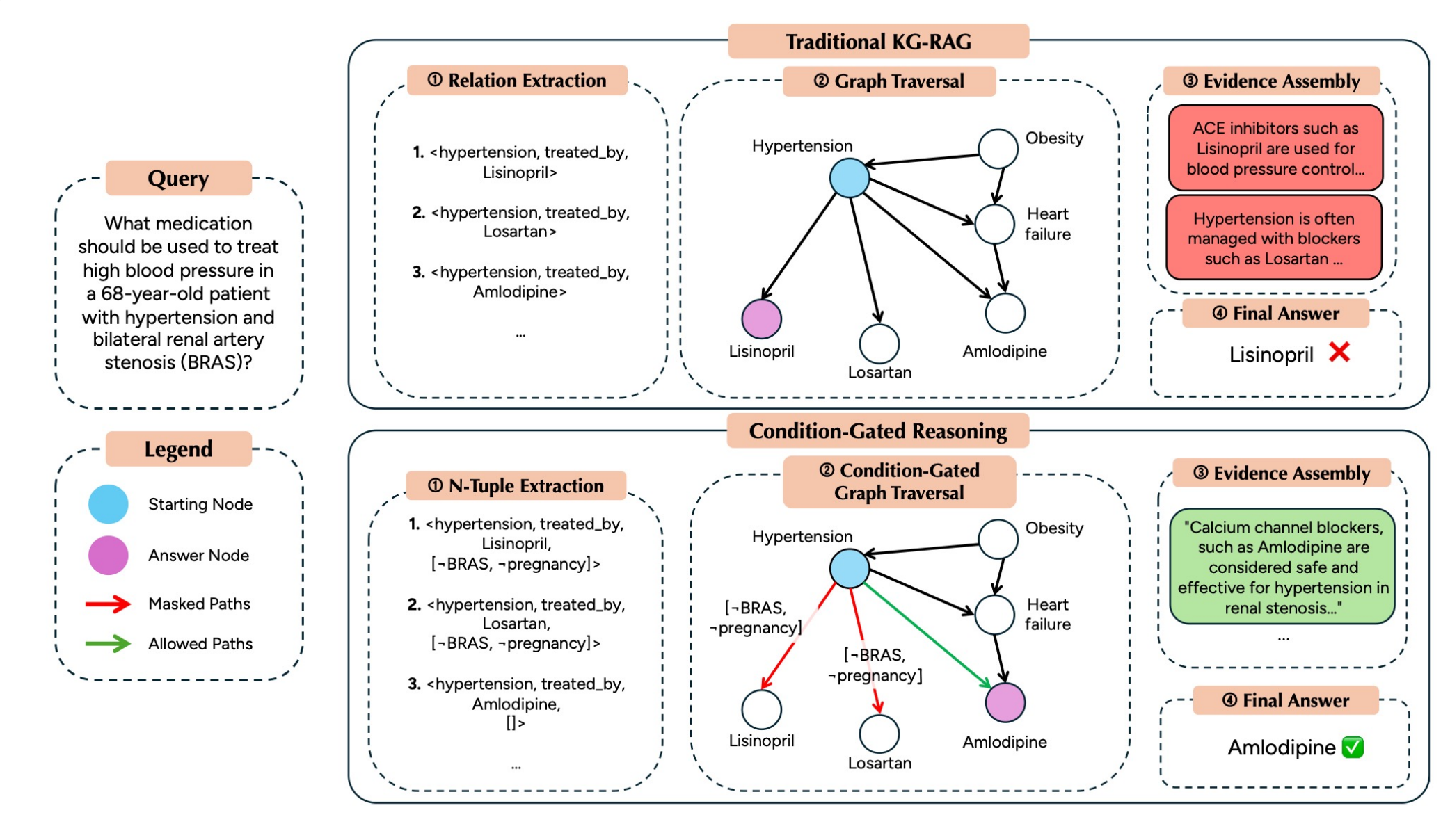
One LLM call checks every condition against the patient context $\rightarrow L: c \rightarrow \{T, F, \emptyset\}$.

Gated traversal

Edge passes if and only if **no** condition is **False**; unknown $\emptyset$ never blocks.

Rank & ground

Score surviving paths; generate an answer traceable to evidence.



Worked example — a hypertension patient with bilateral renal artery stenosis (BRAS). Contraindicated paths (Lisinopril) are **masked**; only condition-safe evidence reaches the **Amlodipine** answer.

State-of-the-Art Performance

+20

EM on conditional QA vs. best baseline

+11

EM on MedHopQA multi-hop

4/4

SOTA on all benchmarks

TABLE 3 · GPT-5.2 · EXACT MATCH / F1

Method	CondMedQA		MedHopQA		MedHop (C)		BioASQ-B	
	EM	F1	EM	F1	EM	F1	EM	F1
Zero-Shot	22.0	41.3	20.5	47.9	28.3	53.5	13.0	40.2
RAG	44.0	52.4	51.3	62.6	45.7	55.0	31.8	54.8
CoT	40.0	52.8	49.5	57.6	50.0	59.4	25.6	47.8
HippoRAG2	46.0	67.6	45.0	60.8	48.6	72.8	22.6	43.6
HyperGraphRAG	57.0	64.5	37.5	41.9	54.3	65.2	28.6	54.4
StructRAG	62.0	71.2	71.8	79.6	65.7	78.6	31.8	55.1
PathRAG	53.0	66.5	30.0	34.4	42.9	54.1	24.6	51.0
MedRAG	57.0	70.5	75.8	84.4	74.3	89.3	35.4	55.5
MKRAG	60.0	71.1	51.3	65.6	62.9	79.3	28.8	55.4
CGR (ours)	82.0	86.7	86.8	89.9	80.0	85.4	40.6	58.0

Gains hold across foundation models — CGR also leads with **Qwen2.5-14B** and **LLaMA-3.1-8B**.

CondMedQA Benchmark

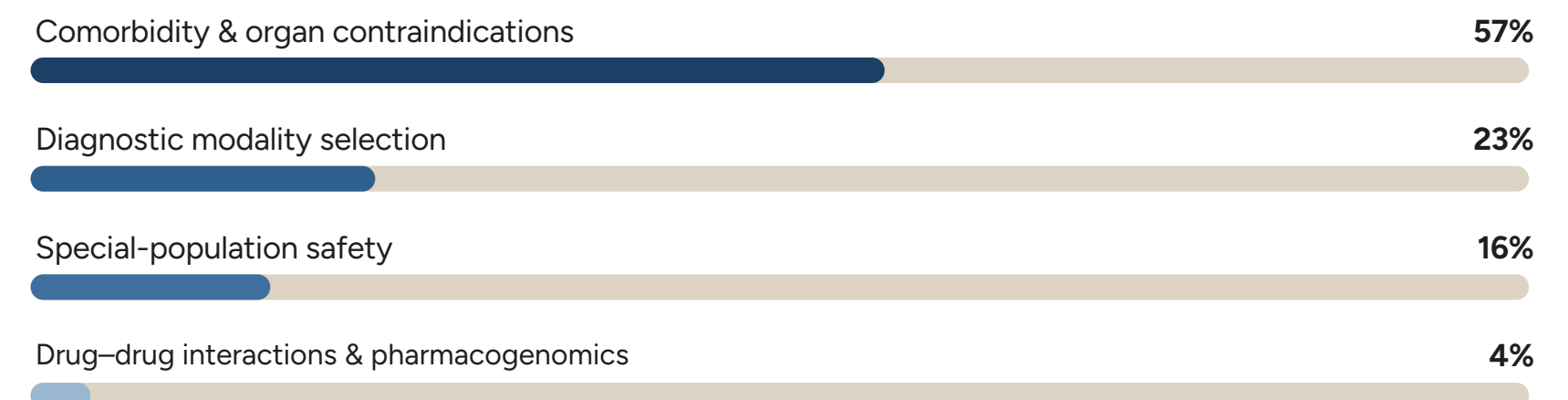
100

verified questions

2-hop

multi-document

REASONING TYPES



WORKED EXAMPLE · SPECIAL-POPULATION SAFETY

Doc 1: First-line antibiotics for Lyme disease: **doxycycline**, amoxicillin, cefuroxime.

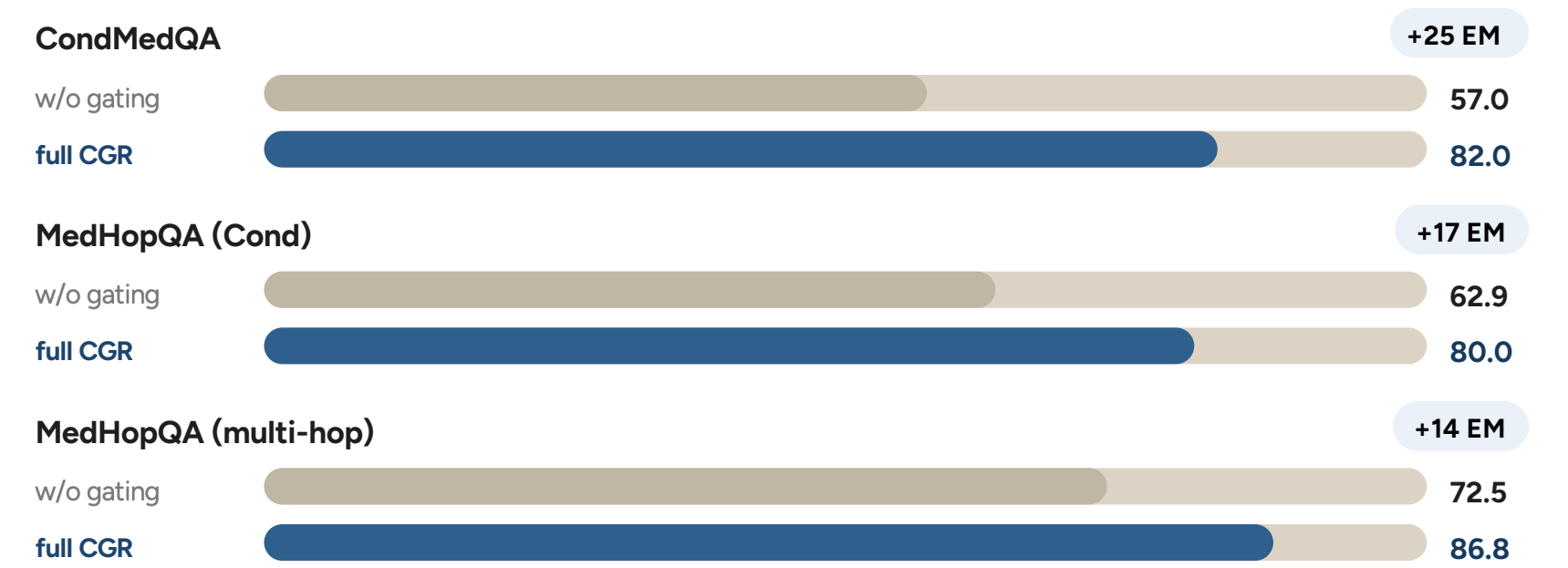
Doc 2: Doxycycline is contraindicated in pregnancy; amoxicillin is safe across all trimesters.

Q: What antibiotic is recommended for Lyme disease in a pregnant patient?

Default: **Doxycycline** $\rightarrow$ Conditional: **Amoxicillin** ✓

Ablation & Analysis

Gating is the driver. Removing the gate — same extraction, graph and traversal — collapses conditional accuracy.



Model-agnostic

A **3B** extractor matches a **14B** one on conditional QA — gains come from gating, not scale.

Robust to Hyperparameter Selection

Scores stay flat across k_{paths} and k_{nodes}

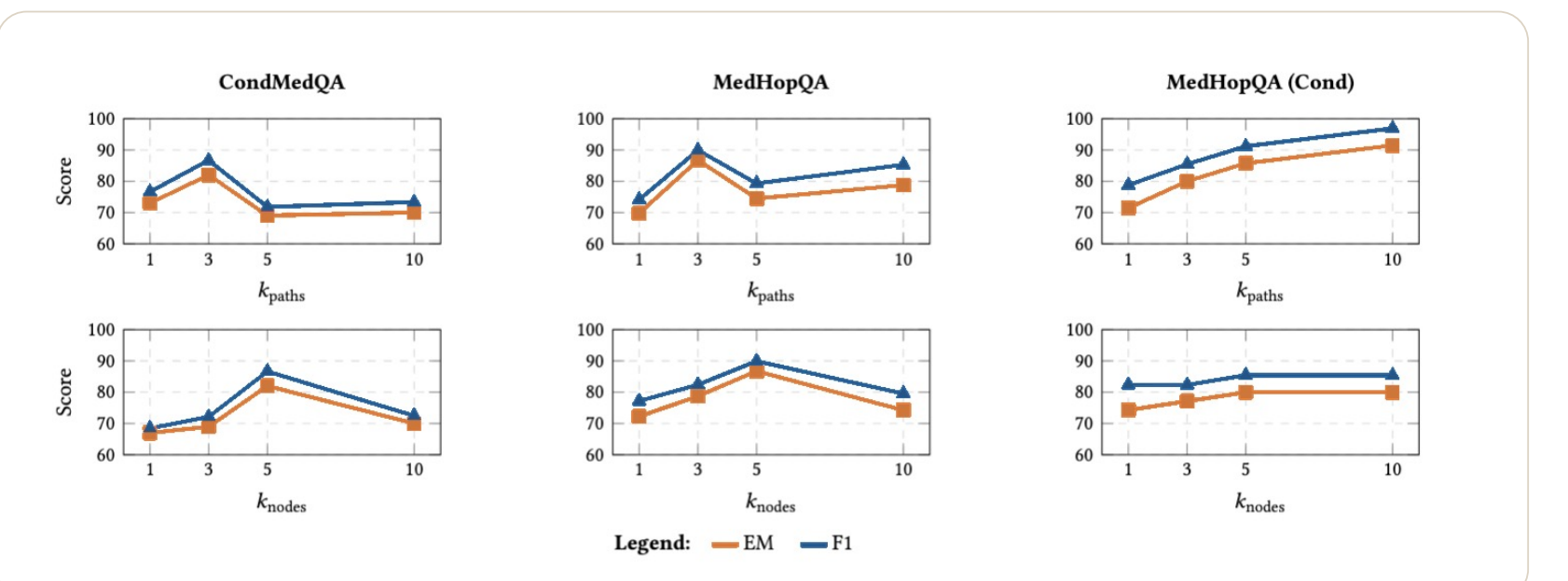


Figure 3 — accuracy stays flat across k_{paths} and k_{nodes} ; CGR is compatible with multiple hyperparameter settings.

KEY TAKEAWAY

Model conditionality **explicitly** — don't treat retrieved knowledge as uniformly valid.

Explore the Work



PAPER



BENCHMARK

Contact

Jash Parekh
jashrp2@illinois.edu